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### Cuneiform and the Rise of Early Alphabets in the Greater Arabian Peninsula: A Visual Investigation

Saad D. Abulhab

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# Cuneiform and the Rise of Early Alphabets in the Greater Arabian Peninsula: A Visual Investigation

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Scholars trace the roots of most historical and modern alphabets in the Near East and Europe, including Arabic and Latin, to a single obscure script, namely the Proto-Sinaitic or Proto-Canaanite script. This presumed script was attested by Western scholars in the early 20<sup>th</sup> Century following the discovery in 1905-06 of a few, very short graffiti inscriptions at “Serabit el-Khadim” in the Sinai Peninsula and the subsequent discovery in 1999 of a few similar ones at “Wadi el-Hol” in the middle of Egypt. According to these scholars, Proto-Sinaitic was derived from the Egyptian Hieroglyphs writing system between 18<sup>th</sup>-15<sup>th</sup> Century BCE and was in use for centuries before the rise of the two major ancient alphabets in the adjacent greater Arabian Peninsula, Phoenician (13<sup>th</sup> Century BCE), and Musnad (9<sup>th</sup>-7<sup>th</sup> Century BCE).

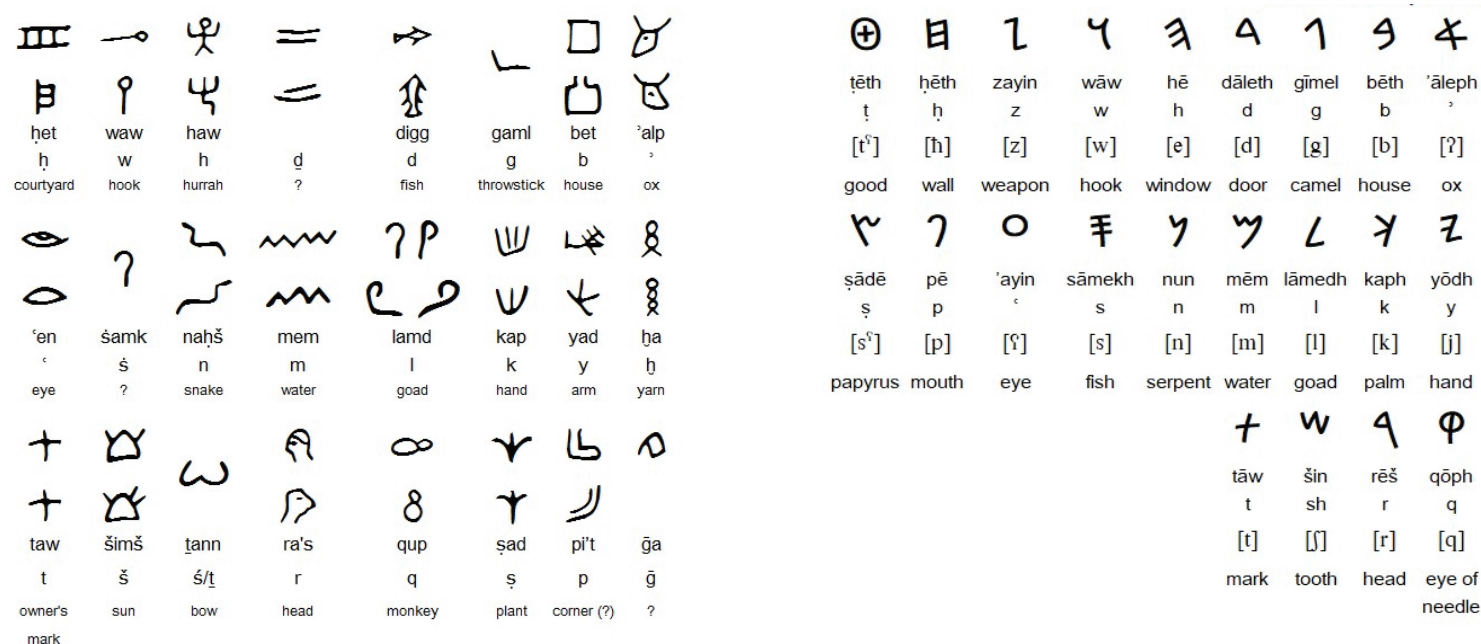


Figure 1: Traced Proto-Sinaitic characters (left) compared to Phoenician characters (Source: Omniglot: The online Encyclopedia of Writing Systems and Languages. <http://www.omniglot.com>)

Suggesting this Egypt-originated Proto-Sinaitic script was the parent script of Musnad and Phoenician is not a completely far-fetched hypothesis, but the presumed starting dates of the two derived scripts is, as we shall see. Surely, the inscriptions from the three scripts do share few letters with similar, derivable shapes (See Figure 1). Furthermore, the well-established “alphabetic” tradition of the ancient Egyptian Hieratic script (30<sup>th</sup> Century BCE) seems to be the most appropriate incubate environment for the rise of the so-called proto-Sinaitic script and the two presumed derived alphabets. This cursive, right-to-left script, featuring abstract shapes mimicking Hieroglyphs, was heavily used, both on stone and papyrus inscriptions, until at least the 3<sup>rd</sup> Century BCE (See Figure 2). Compared to the dominant writing system used at the same period in the Fertile Crescent and beyond, the Mesopotamian Cuneiform system, Hieratic used simpler and more flexible shapes that can easily be identified with those of most historical and modern alphabets.

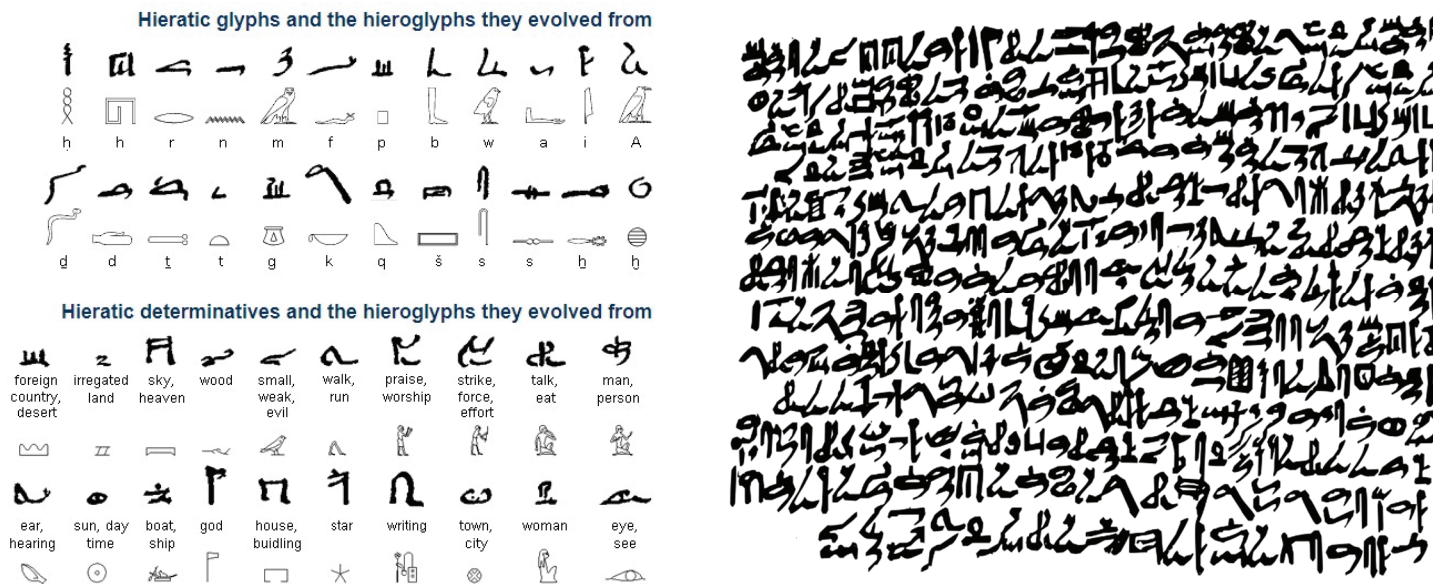


Figure 2: Left: Traced Egyptian Hieratic characters (Source: Omniglot: The online Encyclopedia of Writing Systems and Languages. <http://www.omniglot.com>). Right: Cursive text sample in Hieratic script from Prisse Papyrus (~2000 BCE), the earliest known document on papyrus (Source: Wikipedia. <https://en.wikipedia.org/wiki/Hieratic>)

While Phoenician was a well-defined, single alphabet with 22 letters, Musnad, included at least four flavors, Şafawī, Thamūdī, Liḥyānī, and Saba’ī, each having around 28 letters. These ancient Arabic Musnad alphabets used few significantly different shapes for several exact letters, indicating that they have not been derived from each other only, but from another reference script. The highly advanced and mature Saba’ī Musnad of Yemen, which scholars theorized was a direct child script of proto-Sinaitic, does not seem to be the oldest of the four.

Classifying Musnad as an umbrella script with four slightly differing variants rather than one script, the so-called Southern Arabian Musnad script, is not an arbitrary classification. The four Musnad alphabets have similar shapes for most of their letters. They were utilized by a homogeneous, migrating group of peoples occupying a vast geographic homogeneous area. While the southern Arabian Sabaʿī Musnad script seems more developed or defined visually because it was utilized by the significant civilizations of Yemen, the other Musnad variants, the northern Arabian Ṣafawī and Liḥyānī, and the nomadic Thamūdī were less developed.

The small corpus of inscriptions attested by scholars to prove the existence of a hypothetical proto-Sinaitic script represents rather scant epigraphic evidence. The only inscriptional evidence presented, that of the “Serabit el-Khadim” and “Wadi el-Hol” inscriptions, included a total of only 4 distinguishable short lines sharing a small set of matching shapes. To date, only one line of these 4 text lines is deciphered to some degree of certainty. These graffiti-like inscriptions were found within close vicinity to sites containing many high-quality Hieroglyph inscriptions, indicating they were most likely just a few isolated scrabbles, possibly sketched by non-native workers, or illiterate natives. Theorizing that these few questionable inscriptional sketches represent a solid evidence for the existence of an early script that was utilized for centuries is rather unrealistic and far-fetched assumption.

According to the current timelines suggested by scholars, the parent Proto-Sinaitic script was in use for 2-5 centuries before Phoenician was first derived, and for 6-11 centuries before Musnad was later derived. Scholars claim Sabaʿī Musnad was derived from Proto-Sinaitic 4-6 centuries after Phoenician. This would mean Proto-Sinaitic co-existed with Phoenician for 4-9 centuries! Although old and new scripts can co-exist for long times, there is no single evidence of the existence of Proto-Sinaitic outside Egypt to support the above theory and timelines. Most importantly, after examining the common shapes between Proto-Sinaitic, on one hand, and Phoenician and Musnad, on the other hand, one can logically conclude that the latter two scripts could not have been independently derived from Proto-Sinaitic, as we shall see later.

Scholars often point to the existence of a hypothetical proto-Canaanite script acting as the Proto-Sinaitic precursor of Phoenician that was supposedly used in Canaan, the ancient southern Levant region. They point out to the discoveries of bits and pieces of a few scattered short inscriptions as evidence. However, the inscriptional evidence fails to substantiate this proto-Sinaitic hypothesis. In fact, the most recently found proto-Canaanite inscription, which was discovered in Jerusalem by Israeli archaeologist Eilat Mazar in 2012, pointed to a Musnad origin!<sup>1</sup> The few traced letters in the inscription of the badly broken jar presented by Mazar clearly belong to the northern Arabian Ṣafawī Musnad script, or even Thamūdī Musnad. Unlike the hypothetical proto-Canaanite script, these two scripts are well-documented through the evidence of numerous inscriptions from the historical population centers of the Arabian Peninsula south of the Levant.

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<sup>1</sup> Eilat Mazar, David Ben-Shlomo and Shmuel Ahituv, “An Inscribed Pithos from the Ophel, Jerusalem,” *Israel Exploration Journal* 63 (2013), pp. 39–49.

A close visual examination of Phoenician and Musnad does reveal that they are very likely sister scripts, as scholars theorized, but it does not reveal that they share significant visual characteristics with proto-Sinaitic. Out of the 22 Phoenician letters, only 4, those for Tā', 'Ayn, Jīm, and possibly Ḥā', share sufficient visual characteristics with their presumed corresponding proto-Sinaitic letters. Visually, more than 75% of the Phoenician letters could not have been derived from proto-Sinaitic. Similarly, out of the 26-28 letters in each Musnad script variant, only 4 letters share enough visual characteristics with their presumed corresponding proto-Sinaitic letters. Visually, more than 80% of the Musnad letters could not have been derived from proto-Sinaitic. On the other hand, 13 out of 22 Phoenician letters, those for Alīf, 'Ayn, Wāw, Fā', Shīn, Zā', Tā', Dāl, Nūn, Lām, Jīm, Qāf, and Ḥā', are visually identifiable with Musnad. With more than 50% of its letters compatible with Musnad, Phoenician is clearly a sister script, sharing a single identical root.

Curiously, the four shapes of Phoenician and Musnad sharing clear visual characteristics with proto-Sinaitic belong to the same exact letters in both scripts. At least in term of timelines, this suggests the sister scripts could not have been independently derived, or otherwise developed, from Proto-Sinaitic many centuries apart. It would be an impossible coincidence that the four Musnad variants would solely borrow the shapes of the exact four Proto-Sinaitic letters that Phoenician borrowed 6-11 centuries earlier! A logical conclusion would be that the Phoenician and Musnad scripts had either started independently by like-minded people within a relatively short time period or as one script that eventually branched into multiple ones.

Considering its many shared shapes with Musnad, Phoenician can possibly be classified as another geographic-specific Musnad variant having reduced number of letters and a single, right-to-left order. The large number of Musnad inscriptions discovered all over the Arabian Peninsula (more than 95,000 inscriptions), compared to less than 50 Phoenician inscriptions found in the much smaller adjacent northern area of ancient Phoenicia, indicates Musnad was clearly the predominant script of Arabia. Presuming Phoenician was introduced completely independent of Musnad centuries earlier goes against the facts of history, geography, and visual characteristics. The ancient Levant, like ancient Mesopotamia, was always interconnected with the rest of the Arabian Peninsula and was the destination of many northward migrations and invasions.

However, examining its unique functional and visual features and its distinctive script environment, strongly suggests that Phoenician was most likely developed locally. It seems that Phoenician had borrowed its two most important defining features from the highly successful Ugaritic abjad alphabet, which was the writing system of an adjacent northern Levant region for at least one century earlier. The first borrowed feature was its implementation of a reduced, 22 letters alphabet, which was identical to that of the Ugaritic short alphabet. The Ugaritic long alphabet included 30 letters.<sup>2</sup>

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<sup>2</sup> Abulhab, Saad D. "Roots of Modern Arabic Script: From Musnad to Jazm." Sawt Dahesh. 50-51 (2007-2009)

The second feature borrowed by Phoenician from the Ugaritic alphabet can evidently be seen through its utilization of letter shapes inspired by, and adapted to, the Mesopotamian Cuneiform writing system. Ugaritic conceptualized simplified letter shapes utilizing reduced Cuneiform wedges, while Phoenician conceptualized easy-to-remember abstract shapes based fully or partially on the overall shapes of specific, sound corresponding Mesopotamian Cuneiform syllables. Like the Cuneiform shapes, and unlike the cursive Hieratic ones, these shapes were fully isolated. Even though the Ugaritic alphabet was the predominant script of the city/state of Ugarit, modern day “Rās Shamrah” in Syria, the Akkadian Cuneiform (~3500 BCE) was still the predominant script of the Levant and Mesopotamian, including ancient Phoenicia.

Examining the Phoenician shapes, at least 6 letters, those for ‘Ayn, Wāw, Bā’, Rā’, Sīn, Shīn, and Lām can easily be linked to the Akkadian Cuneiform symbols for the e, u, ba, ra, sí, and lá syllables, respectively. Six more shapes, those for Yā’, Tā’, Dāl, Nūn, Jīm, and Kāf can also be linked to the i, ta, da, nu, gi, and qa syllables, respectively. Creating abstract Cuneiform shapes mimicking drawings was how the early few Mesopotamian cuneiform symbols developed. In adjacent Egypt, the abstract shapes of the letters and syllabus of the much older and advanced Hieratic Script were also based on the pictorial Hieroglyph drawings. Most likely, the Phoenician script was invented by utilizing the Ugaritic approach to create a small alphabet based on Cuneiform and employing the Egyptian precedent of creating simpler abstract shapes corresponding to those of a well-established writing system to be easily scribed, memorized and linked to the established script.

Although the Mesopotamian Cuneiform was still the dominant writing system east of Egypt, and was in use through the early centuries CE, it seems that it was gradually being replaced by various reduced, visually simplified alphabets, at least since the rise of the Ugaritic around the 14<sup>th</sup> Century BCE. While Ugaritic, Phoenician, and the four Musnad variants were developed to replace the Cuneiform script, they were used simultaneously with it, for a long time. A new writing system designed to replace an old one often preserves some of its visual and functional features. It is often developed with keeping the old writing system in mind. The new is typically born within the old. The new rarely obliterates the old. The Mesopotamian Cuneiform writing could not, and did not, vanish without trace. During the long period of its gradual disappearance, it was clearly the reference point in a complex evolutionary process leading to the rise of several key alphabets in the region.

Examining the letters of the four Musnad variants (and to a lesser extend the Phoenician) clearly reveal most of their shapes can be visually traced to their sound-corresponding Akkadian Cuneiform syllable symbols. The Musnad abstract shapes depicted, partially or fully, the overall look and feel of a small subset of Cuneiform symbols, employing one or more processes including elimination, enclosing, flipping, rotating, mirroring, shrinking, or extending. The accent basis used to select these syllables strikingly matches the accent basis Arabic used historically to name its letters (i.e. Nūn vs. nu, or Lām vs. la). The different shapes for a few identical letters across the Musnad variants, and even Phoenician, represent either different abstract shapes of identical similar-sounding Cuneiform syllables, or abstract shapes of different, similar-sounding corresponding Cuneiform syllables. Except for the letters Yā’, Khā’, and Qāf, all Musnad shapes can be identified with Akkadian Cuneiform.

It is difficult to conclude with complete certainty whether Phoenician did exist for centuries before any Musnad variant existed, as scholars claim today. Having a handful of older Phoenician inscriptions is not a decisive factor. Determining inscription dates is not a precise science. Except when explicitly stated in the inscriptions, dates cannot be concluded with absolute certainty. Even when an older date is accurate beyond doubts, it does not indicate the age of the script used because one cannot exclude the existence of undiscovered earlier inscriptions.

Our visual analysis of Phoenician indicates it was at least partially derived from the Musnad Ṣafawī script. The fact that Phoenician was a well-developed script with a clearly defined direction, compared to Musnad, further indicates it is was not older. The Phoenician letters Alīf and Qāf seem to be derived from the Ṣafawī Musnad shapes. In both letters, the three strokes used to form the Musnad shapes were reduced to two in the corresponding Phoenician shapes! Also, the Phoenician shape for the letter Dāl seems to be derived from the Musnad Ṣafawī Dāl, reducing two strokes to one. In all three cases, it is impossible to visually link these Phoenician shapes to proto Sinaitic!

Additionally, the shapes of some Phoenician and Musnad letters seem to be derived from the shapes of their language-based, alternative sounding letters within each script. For example, in Phoenician, the shapes of the letters Hā', Mīm, and Ṭā' seem to be derived from the shapes of their alternative letters Ḥā', Nūn, and Tā', respectively. In Saba'ī Musnad, the shapes of the letters Hā' and Khā' seem to be derived from the shape of letter Ḥā', which was clearly evolved from an equivalent letter shape in the Ṣafawī Musnad script. The shapes for the letters Ṣād, Ṭā', Ṣā' in the Saba'ī Musnad also seem to be derived from each other over time. The shapes of the letters Fā' and Bā', which are historically interchangeable in Arabic, have very similar shapes in Musnad Liḥyānī.

Similarly, in the Akkadian Cuneiform system we see single symbols representing multiple interchangeable syllables. For example, a single Cuneiform symbol, that for the syllable ḥa, was used to represent the equivalent of the Arabic letters Hā', Ḥā', and Khā'. Similarly, a single Cuneiform symbol, that for the syllable ša, was used to represent the equivalent of Arabic letters Thā' and Dhāl. Earlier on, during the rise of the Akkadian writing system from Sumerian, the two syllables ba and fa, the two syllables si and ši, the three syllables sa, sa, and za, and the four syllables ta, da, ša, and ṭa, also shared a single Cuneiform symbol in each case.

The observations outlined in the above two paragraphs should not come as a surprise to anyone. The three writing systems, Cuneiform, Musnad, and Phoenician, were primarily used to record a single evolving language, namely the early Arabic language! The non-existing, unattested "Semitic" language, which was coined in the early 19<sup>th</sup> century based on biblical teachings is intentionally ignored in this study.<sup>3</sup> In fact, the exact interchangeable letters that were specified in the two paragraphs are well-documented in the historical Arabic language manuscripts. The

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<sup>3</sup> Abulhab, Saad D. DeArabizing Arabia: Tracing Western Scholarship on the History of the Arabs and Arabic Language and Script. New York: Blautopf Publishing. 2011

breakthrough translations by the author of more than 1400 text lines from two major Akkadian tablet-inscribed documents, using the historical Arabic etymological references, clearly showed that the language of these documents was indeed early classical Arabic.<sup>4</sup>

Linking Musnad and Phoenician to Cuneiform, rather than Proto-Sinaitic, is not a stretched hypothesis by the author. Just to point out few, the reader should carefully examine the following striking visual similarities. Even in a relatively late Saba'i Musnad inscription from Yemen, one can clearly recognize the Cuneiform wedges forming many of its letters shapes. The fifth column of the table below shows isolated letter images from the Sa'adTa'lib Musnad inscription, which was originally read by the author in 2011.<sup>5</sup> Notice in that table how the Musnad Liḥyānī letter Dhāl is identical to the Akkadian Cuneiform symbol for the Syllable šá, and the Musnad Šafawī letter Ḥā' is identical to the Akkadian Cuneiform symbol for the Syllable ha. Also notice how the Phoenician letter Sīn is identical to the Akkadian Cuneiform symbol for the Syllables sí or sé, and the letter Kāf is identical to the Akkadian Cuneiform symbol for the Syllable qa. The letters Qāf and Kāf were interchangeable in Sumerian and Akkadian. Again, in all these examples, it is impossible to visually link these shapes to Proto-Sinaitic!

Western scholars' hypothesis about Musnad and Phoenician evolving from Proto-Sinaitic is usually supported by their linkage of letter shapes/names to objects and drawings (i.e. Phoenician Aleph, or Alif, for "ox" in Arabic). This, however, is misleading. Even though most Proto-Sinaitic shapes clearly mimic objects and drawings and could have been referred to accordingly, most Phoenician shapes hardly resembled objects drawings and their presumed objects/names did not correspond to the Proto-Sinaitic ones. For example, the proto-Sinaitic presumed shapes/names for the letters Jīm, Dāl, Hā', Fā', Qāf, Ḥā', and Shīn represented the drawings/names of "throw stick", "fish", "hurrah", "corner", "monkey", "courtyard", and "tooth", respectively, but the corresponding Phoenician shapes/names supposedly represented the drawings/names of "camel", "door", "window", "mouth", "monkey", "eye of needle", "wall", and "sun", respectively. Like the Akkadian Cuneiform symbols, most Arabic Musnad shapes cannot be visually linked, with any degree of certainty, to objects drawings. Incidentally, most Arabic letters names were clearly invented to identify syllable sounds, not objects.

The objects/names discrepancies outlined above further indicate that the Phoenician shapes have not evolved from the proto-Sinaitic ones. The letter names of both Phoenician and Musnad (if any) were likely assigned according to one of the following three reasons: to link letter sounds with names of objects starting with them (i.e. Sheen as-in shin, the Northern Arabic word for "tooth"), to link already developed shapes abstracts to objects resembling them (i.e. Phoenician letter Qoph shape resembles "qawf", Arabic for "eye of a needle"), or to assign independent, sound-based syllable names (i.e. Shīn for the syllable sound shi). It is important to point out here that we do know how the Proto-

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<sup>4</sup> Abulhab, Saad D. *The Law Code of Hammurabi: Transliterated and Literally Translated from its Actual Early Classical Arabic Language*. New York: Blautopf Publishing. 2017; Abulhab, Saad D. *The Epic of Gilgamesh: Selected Readings from its Original Early Arabic Language*. New York: Blautopf Publishing. 2016.

<sup>5</sup> Abulhab, Saad D. *Inscriptional Evidence of Pre-Islamic Classical Arabic: Selected Readings in the Nabataean, Musnad, and Akkadian Inscriptions*. New York: Blautopf Publishing. 2013.



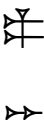
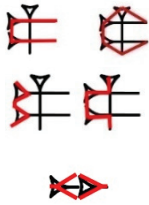
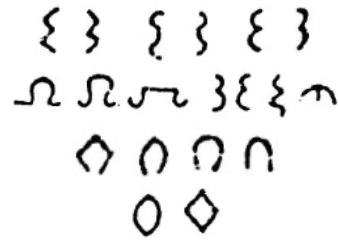
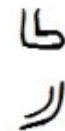
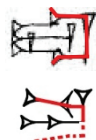
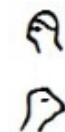
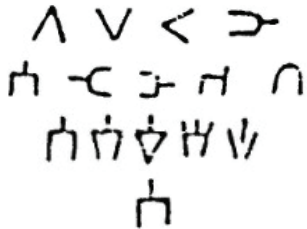
Sinaitic letters were named, if ever. If any, the minor overlapping of shapes/names between Phoenician, Musnad and proto-Sinaitic could indicate that the scribes of the handful Proto-Sinaitic graffiti inscriptions were previously exposed to both Musnad and Phoenician and were simply attempting to adapt to a new, drawing-based script environment. There is no solid evidence to prove otherwise.

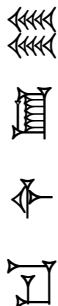
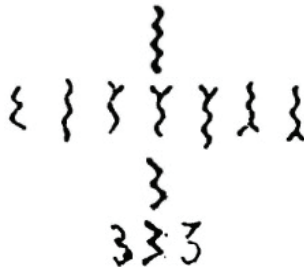
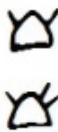
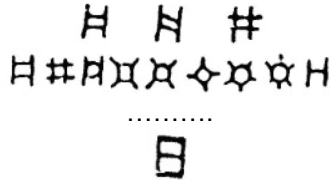
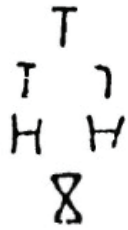
In conclusion, this study primarily emphasizes shapes and scripts functional characteristics analysis. After visually analyzing the Proto-Sinaitic, Musnad and Phoenician shapes, the author believes there is no solid evidence to conclude that Proto-Sinaitic was the parent script. The author further believes that Musnad and Phoenician started either as independent scripts over a relatively short time, or as one script splitting into multiple ones over a relatively long period, to replace and visually preserve, at the same time, a predominant, concurrently-used older writing system: namely the Akkadian Cuneiform script. In either scenario, these early alphabets represented different iterations to mimic the Akkadian Cuneiform script, abstracting a selective set of syllables into simpler, easier to handle and remember linked shapes. Visual evidence reveals Cuneiform was their model parent script for a long period.

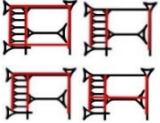
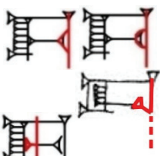
In the following table, readers can examine the visual observations pointed out by the author and/or conclude their own. This table compares shapes from the four variants of Musnad, Phoenician, and the so-called Proto-Sinaitic scripts, side by side shapes from their corresponding Akkadian Cuneiform syllables. For Cuneiform, the table uses shapes from a typical modern font in addition to actual, traced shapes from the stele of the Code of Hammurabi. For Musnad, the table uses pictures of isolated shapes from the Saba'i Musnad Sa'adTa'lib inscription stone. Using red color, the author put forth a few scenarios of how the Musnad and Phoenician shapes could have abstracted the Akkadian Cuneiform syllables or evolved from each other.

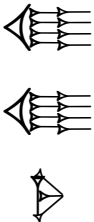
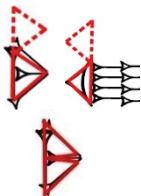
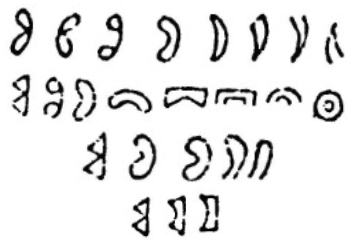
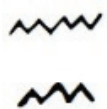
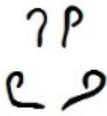
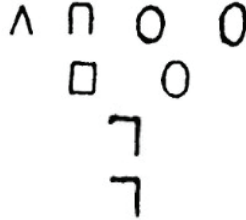
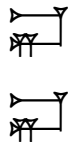
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|                            | Font               | Hammurabi<br>Stele<br>Tracings | Transformations | Images from<br>Sa'adTa'lib | Şafawī, Thamūdī, Liḥyānī, Saba'ī<br>Tracings<br>(Top to Bottom, Respectively) | Tracings (Top)<br>Transformations<br>(Bottom) | Two<br>Tracings |                            |
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| pa<br>pá=BA                            |    |                                                                                    |    |    |    |   |  | ف |
| ra<br>rá=DU<br>řá=DU                   |    |   |    |    |    |   |  | ر |
| sà=ZA<br>se=SI<br>sé=ZI<br>si<br>sí=ZI |   |  |   |   |   |  |  | س |
| šá                                     |  |                                                                                    |  |  |  |                                                                                      |                                                                                     | ذ |

|                             |                                                                                     |                                                                                   |                                                                                     |                                                                                   |                                                                                       |                                                                                       |                                                                                     |   |
|-----------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---|
| še<br>še, šè<br>ši<br>ší=SI |    |                                                                                   |    |  |    |    |  | ش |
| ša                          |    |  |    |  |    |    |  | س |
|                             |                                                                                     |                                                                                   |                                                                                     |  |   |                                                                                       |                                                                                     | ض |
| sà=ZA<br>za<br>zá           |  |                                                                                   |  |                                                                                   |  |  |                                                                                     | ز |

|                   |                                                                                     |                                                                                    |                                                                                     |                                                                                     |                                                               |                   |                   |   |
|-------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------|-------------------|---|
|                   |                                                                                     |                                                                                    |                                                                                     |    | <p>Მ Ლ Მ Მ Მ Მ Მ</p> <p>.....</p> <p>.....</p> <p>Ი Კ Მ Მ</p> |                   |                   | ظ |
| ta                |    |   |    |    | <p>Მ Მ Მ Მ Მ</p> <p>Მ Მ Მ Მ Მ</p> <p>Მ</p> <p>Მ Მ</p>         | ⊕                 | ω                 | ظ |
| ta<br>tá=DA<br>ša |    |   |    |    | <p>Ი Მ</p> <p>Ი Მ</p> <p>Ი</p> <p>Ი</p>                       | +                 | +                 | ظ |
| da<br>dá=TA       |   |  |   |   | <p>Კ Მ Კ Მ Კ Მ</p> <p>Კ Კ Მ Მ Მ</p> <p>Კ Კ Კ Კ</p> <p>Კ Კ</p> | <p>Δ</p> <p>Კ</p> | <p>→</p> <p>↗</p> | ظ |
| nu                |  |                                                                                    |  |  | <p>Მ</p> <p>Მ Მ Მ Მ Მ</p> <p>Მ Მ Მ</p> <p>Მ Მ Მ</p>           | γ                 | <p>~</p> <p>~</p> | ظ |

|                            |                                                                                     |                                                                                   |                                                                                     |                                                                                     |                                                                                                |                                                                                              |                                                                                              |          |
|----------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------|
| <p>mé=MI<br/>mi<br/>mí</p> |    |                                                                                   |    |    | <p>  </p>   | <p>  </p> | <p>  </p> | <p>م</p> |
| <p>la<br/>lá</p>           |    |                                                                                   |    |    | <p>  </p>   | <p>  </p> | <p>  </p> | <p>ل</p> |
| <p>gi<br/>gí<br/>ge=GI</p> |    |  |    |                                                                                     | <p>  </p>   | <p>  </p> | <p>  </p> | <p>ج</p> |
| <p>gá<br/>gǎ</p>           |  |                                                                                   |  |  | <p>  </p> |                                                                                              |                                                                                              | <p>غ</p> |

|                |  |  |  |  |  |  |  |   |
|----------------|--|--|--|--|--|--|--|---|
| ka<br>ká<br>kà |  |  |  |  |  |  |  | ك |
| qa             |  |  |  |  |  |  |  | ق |
| ba<br>bà       |  |  |  |  |  |  |  | ب |
| kha            |  |  |  |  |  |  |  | خ |
| ha             |  |  |  |  |  |  |  | ه |